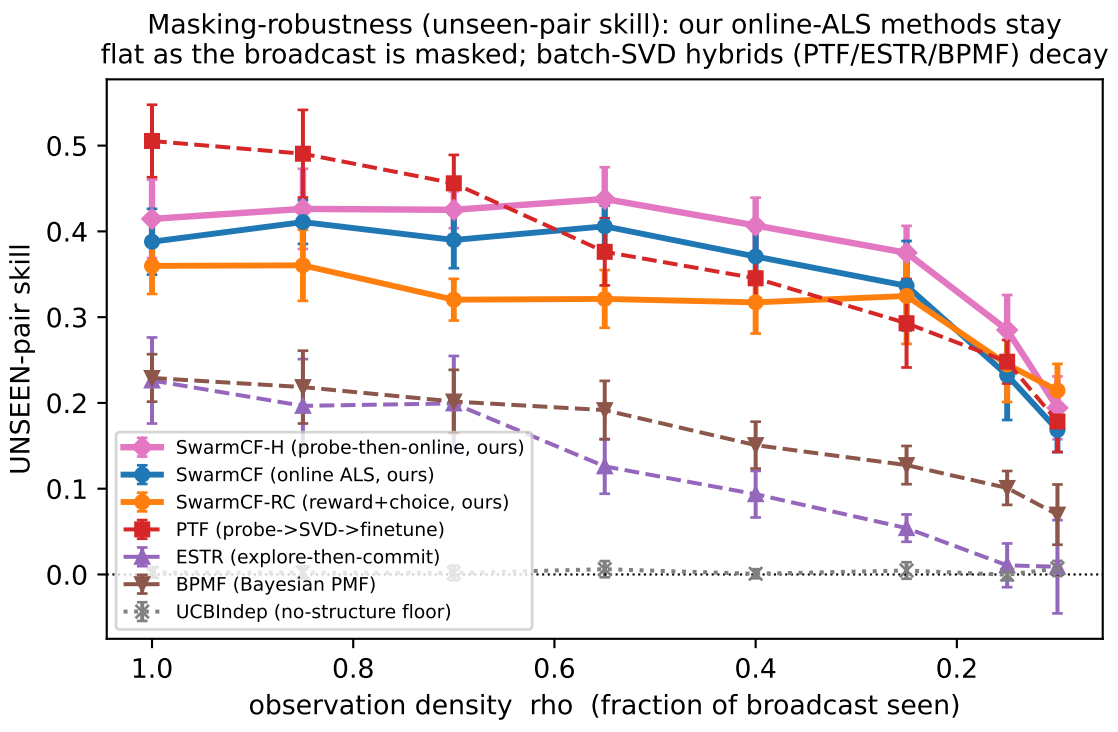
Masking-robustness (unseen-pair skill): our online-ALS methods stay flat as the broadcast is masked; batch-SVD hybrids (PTF/ESTR/BPMF) decay
UNSEEN-pair skill
0.5
0.4
0.3
0.2
0.1
0.0
1.0
0.8
0.6
0.4
0.2
observation density rho (fraction of broadcast seen)
SwarmCF-H (probe-then-online, ours)
SwarmCF (online ALS, ours)
SwarmCF-RC (reward+choice, ours)
PTF (probe->SVD->finetune)
ESTR (explore-then-commit)
BPMF (Bayesian PMF)
UCBIndep (no-structure floor)